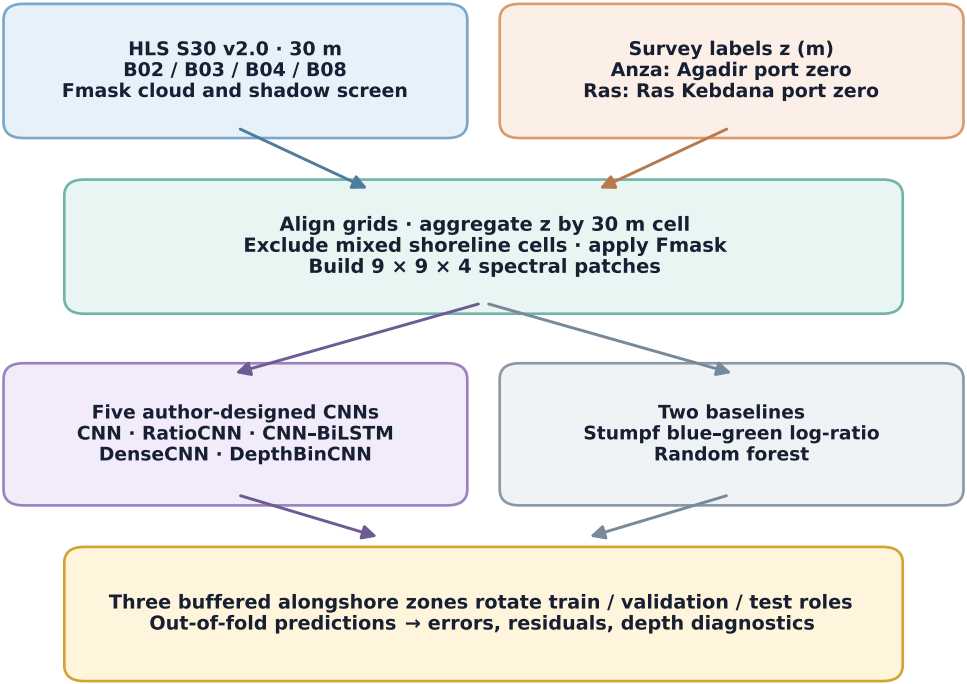
Shared processing · separate site evaluations
HLS S30 v2.0 · 30 m
B02 / B03 / B04 / B08
Fmask cloud and shadow screen
Survey labels z (m)
Anza: Agadir port zero
Ras: Ras Kebdana port zero
Align grids · aggregate z by 30 m cell
Exclude mixed shoreline cells · apply Fmask
Build 9 × 9 × 4 spectral patches
Five author-designed CNNs
CNN · RatioCNN · CNN–BiLSTM
DenseCNN · DepthBinCNN
Two baselines
Stumpf blue–green log-ratio
Random forest
Three buffered alongshore zones rotate train / validation / test roles
Out-of-fold predictions → errors, residuals, depth diagnostics
Local vertical zeros stay separate · no cross-site transfer test